Hyper Local Aviation Weather Forecasting and Verification for Sri Lankan Monsoon Conditions

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***Abstract*— Short-term weather forecasting for accurate aviation weather prediction is significant in ensuring safe and efficient operation is significant in ensuring safe and efficient operation of airports especially in Sri Lanka considering that weather in Sri Lanka change in accordance with monsoon seasons. The main seasons taken into consideration in this research include Southwest Monsoon, Northeast Monsoon, First Inter Monsoon, and Second Inter-Monsoon. In this research, Hyper-Local aviation weather forecasting and verification system is introduced for VCBI. Four models including pressure-Humidity and QNH Stability Model are developed to predict aviation weather conditions. Different machine learning algorithms are employed in creating these four models using historical meteorological observations for airports. METAR observation and TAF forecast are used as aviation weather information for the verification process. The proposed system creates three hours ahead weather forecast and verifies the predicted conditions with respect to available weather information in order to determine the accuracy of the predictions.**

Keywords—Aviation Weather, Hyper-Local Forecasting, Machine Learning, Monsoon, Weather Verification, Airport Weather Forecasting, Short-term Prediction

I. INTRODUCTION

The need for reliable and accurate weather data is fundamental in ensuring the safety and efficiency of the flight process due to atmospheric conditions like temperature, pressure, humidity, wind, visibility, and cloudiness, which could affect the operation of flights, use of runways, and other decisions regarding the aerodromes [3], [5]. The changes in these weather elements occur quickly and cloud vary at airports in tropical regions. Consequently, accurate short-range forecasts of weather elements are necessary for aviation operations.

Sri Lanka faces seasonal changes in the atmosphere because of the Southwest Monsoon, Northeast Monsoon, First Inter-Monsoon, and Second Inter-Monsoon seasons. Some studies have indicated the use of Machine Learning (ML) and Artificial neural networks for weather prediction in Sri Lanka such as the monsoon rainfall forecast and short-term weather forecast [1], [2]. In most cases, most of the works are about regional weather behavior, rainfall prediction, and generally short-term forecasting, not hyper-local aviation weather forecast at one particular airport. This is made necessary by the dynamic nature of tropical weather.

Meteorological Aerodrome Report (METAR) and Terminal Aerodrome Forecast (TAF) are two examples that supplement each other in the context of weather in the field of aviation meteorology. The first is weather observations of today at the aerodrome, while the second is a forecast of today at the aerodrome. Some of the recent studies related to aviation meteorology are Artificial Intelligence (AI) for automated TAF generation and Deep Learning (DL) on the forecasting of convective weather in civil aviation operations [6], [13]. Other high-impact weather of aviation involves strong winds, thunderstorms, turbulence, and low visibility [5]. However, this study does not consider the use of airport historic data, monsoon in Sri Lanka, short-term forecasting, and verification of results.

To bridge this gap, the current study will develop and AI-enabled Hyper-Local weather Forecasting and Validation system for Bandaranaike International Airport (VCBI). The proposed system consists of four main prediction models designed to forecast different aviation weather parameters. These include a Pressure-Humidity-temperature Coupled Prediction Model, a Wind Pattern Prediction Model, a Cloud and Visibility Prediction Model and QNH, Dew Point, and Relative Humidity (RH) Prediction Model. These models are developed using different ML approaches according to the characteristics of the respective weather parameters. The proposed models generate short-term airport airport-level weather forecasts, which are subsequently verified using aviation weather forecasts, which are subsequently verified using aviation weather observations to evaluate verified using aviation weather observations to evaluate their prediction performance.

Another significant factor concerning the proposed methodology is the combination of forecasting and verification methods. Short-term forecasts generated through the method are checked against future METAR using different

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error indicators such as Mean Absolute Error (MAE), Root Mean Square Error (RSME), and coefficient of determination (R^2). In addition to this, the corresponding TAF forecasts but also provides an evaluation of the quality of the produced forecast. The aim is to develop and airport-based monsoon weather forecasting tool.

II. Literature Review

A. Machine Learning for Weather Forecasting

ML and AI technologies have recently been used for weather forecasting due to their capability of recognizing complex relationships between variables from past records of meteorological observations. It is especially important for Sri Lanka, which experiences weather variability caused by at atmospheric conditions that result from monsoons. In their work, Nagamulla, Rathnayake, and Ratnaweera [1] examined the application of artificial neural networks for monsoon rainfall forecasting in Sri Lanka based on climatic teleconnections. Hennayake, Dinalankara, and Mudunkotuwa [2] created a ML based short term weather prediction model for Sri Lanka, taking into account weather variables such as temperature and precipitation. All these studies show the weather of Sri Lanka. However, both studies concentrate primarily on weather prediction and do not consider local hyper-weather forecasting at airports.

Future advances in the field of weather prediction have been made due to DL algorithms. Ahi *et al*. [7] suggested using the Convolutional Long Short-Term Memory (ConvLSTM) network for precipitation nowcasting by reformulating it into spatiotemporal sequence forecasting. The results have shown that ConvLSTM is able to consider both temporal and spatial dependencies of precipitation. Some other work has been done on AI and neural networks for real-time weather prediction [8]. Furthermore, Dueben and Bauer [9] described the difficulties and possible solutions of the application of ML in global weather and climate models. Also neural Networks have been considered for postprocessing of ensemble weather predictions [10], which shows that only can machine learning produce forecast itself, but it can also be used to improve the existing forecast.

B. Aviation Weather Forecasting

Aviation Weather forecast requires extra conditions because changing conditions in the atmosphere could affect the operation of the aircraft, the flight plan, and the safety at airports. Oo and Oo [3] have carried out research on aviation weather risks and their underlying causes in the Yangon Flight Information Region. In their study, thunderstorms and fog were identified as major weather risks that impact aviation operations while linking the impact of weather conditions to operations. Likewise, Gultepe *et al*. [5] analyzed high-impact weather in aviation meteorology and pointed out the significance of accurate measurements and nowcasting.

AI has also been studied in relation to aviation forecasting. Sládek [6] proposed an AI-based method for automated TAFs looking into various machine learning techniques that can predict aviation parameters like wind speed and visibility. This research shows how the methods based on AI could be used for automated airport forecast generation.

C. Machine Learning in Aviation Prediction

ML methods used for the analysis of aviation prediction have included techniques such as Random Forest (RF). In a paper by Hu, Zhang, and Li [11], the use of RF in flight delay prediction has been examined, indicating the use of ML in the field of civil aviation. Even though this is different from weather prediction, this study shows the broad uses of RF in aviation predictions.

Furthermore, Boonvuen *et al*. [12] studied the effect of Global Positioning System (GPS) Radio Occultation and Radiance Data Assimilation on numerical weather prediction. This clearly shows how essential it is to use the right data sources in building weather predictions due to the influence of the input in the input in the prediction model.

D. Aviation Weather Hazards and Research Gap

Aviation Weather forecasting entails further requirements as changes in atmospheric conditions can have a direct effect on the operation of aircraft, the planning of flights and airport safety. Wang *et al*. [3] explored aviation weather hazards and the mechanisms behind them within the Yangon Flight Information Region, which shows the significance of taking into account weather conditions at aviation areas specifically. Sprenger and others' work on high-impact weather for aviation meteorology [5] and the need for precise and timely weather information for aviation operations also shows the significance of weather forecasting for aviation. Moreover, AI has been explored for aviation-specific forecasting. Jeong [6] suggested an AI approach to automated (TAF)

Other research associated with aviation prediction concerns ML approaches, including RF. Hu, Zhang, and Li [11] explored the problem of flight delay prediction via the use of RF method, which proves the possibility of applying ensemble ML techniques to the area of civil aviation. Even though flight delay prediction is not similar to weather forecasting, this example can be used as proof of the versatility of RF approach in aviation-related problems. Moreover, Boonyuen *et al*.[12] researched the impact of GPS radio occultation and radiance data assimilation on Numerical Weather Prediction.

Also, weather-related risks related to particular airport climatologies need consideration. In this regard, Jayasinghe [4] explained the climatology of lightning activities over VCBI in Sri Lanka. Recent studies by Asifa *et al*. [13] examined lightning risk prediction for aviation safety, emphasizing lightning risk prediction for aviation safety, emphasizing the importance of weather-related risk assessments for aviation safety_once_again._These_studies show that_specific weather_information_is _necessary_for aviation risks_assessment at airports.

Even though some previous works provide good background for AI-based weather and aviation research, there are still some gaps in these research areas. Works by Sri Lankan researchers deal mostly with general weather forecasting and monsoon rainfall forecasting and monsoon rainfall forecasting [1], [2], whereas aviation-related researches examine weather hazards and automated aviation forecasting [3], [6]. Also, DL and and ML researches demonstrate the value of temporal and spatiotemporal models for weather prediction [7], [9], and atmospheric data assimilation researches prove the necessity of appropriate meteorological observations [12]. However, the reviewed works do not directly address the joint prediction of temperature, atmospheric pressure, and RH at VCBI in the conditions of Sri Lankan Monsoon.

III. METHODOLOGY

A. Overall Research Framework

The methodology is implemented using a modular framework that involves data acquisition, pre-processing, feature engineering, modeling, short term forecasting, physical validation of forecasts, and model maintenance. Past VCBI observations are employed to build the forecasting models, while current METAR observations are used to do real-time inference and validation of forecasts. The key forecasting period is T+3 where the wind forecast is T+1 as well.

B. Dataset and Proprocessing

Past meteorology data obtained from VCBI are primary source of training data. The temperature & pressure model utilizes over 120,000 past data samples for the period 2019-2026. Current METAR weather date are obtained from the Aviation Weather Center Website, extracted, and then stored locally in an SQLite database. Some of the preprocessing techniques used are chronological ordering, deduplication, handling missing data, temporal feature extraction, and dataset preparation for particular models. When necessary, Min-Max normalization is used. The training and test datasets are separated chronologically without shuffling.

C. Feature Engineering

Feature Engineering is done based on the physical and temporal properties of each feature. For the Temperature and Pressure feature, it utilizes lagged meteorological features and 6 hours window for Long Short-Term Memory (LSTM) computations. Wind feature system encodes wind direction through sine and cosine transformation with lagged and momentum features. Cloud and Visibility feature system generates dew point depression, temperature-RH interaction, wind speed-RH interaction, QNH-wind interaction, squaring of RH values, and encoding of present weather conditions. The QNH/Dewpoint feature system incorporates cyclic hour transformation, temperature difference, pressure lags and momentum, and three-hour and six hour rolling averages.

D. Temperature and Pressure Prediction

The temperature and pressure forecasting subsystem uses the hybrid approach based on the combination of RF and LSTM models. The former allows modeling of nonlinear relationship between humidity, dewpoint, wind and lagged features, whereas the latter can capture the dependencies in sequences of meteorological observations. The outputs of both models are combined to provide a temperature forecast at T+3 and QNH forecast (pressure). It was reported that historically the baselines had an R^2 coefficient equal to about 68% for RF 25% for LSTM and 84% for the hybrid approach, and MAE approximately equal to 0.866 for the latter one.

In order to convert these forecasts to an aviation relevant index, the obtained T+3 temperature and QNH predictions were used to obtain Density Altitude (DA). At first, the Pressure Altitude (PA) is computed through the difference between predicted QNH and the standard sea level pressure of 1013.25 hPa:

PA (feet) = (1013.25 - Predicted QNH)* 30

Next, he International Standard Atmosphere (ISA) temperature is obtained using the standard sea level temperature of 15°C and laps rate of 2°C/1000 feet:

ISA Temperature (°C) = 15 -[2 * (PA / 1000)]

The ISA temperature deviation is defined as follows:

ISA Deviation (°C) = Predicted Temperature – ISA Temperature

And finally, the DA is calculated based on the above formula:

DA (feet) = PA + (118.8 * ISA Deviation)

Thus, the hybrid approach of ML along with standard aeronautical physical formulas allowed he conversion of predicted temperature and pressure conditions into aviation relevant DA. The obtained DA at VCBI is categorized as Normal, Elevated and Critical depending on the operational thresholds.

E. Wind Prediction

This particular section elaborates on the method and architecture of the Hyper-Local Wind Prediction and ML Ops Engine, which forms the core of the overall predictive module. The main goal of this particular module is to predict volatile and runway-specific wind pattern in the VCBI. This is achieved through the integration of the continuous cloud-based pipeline, advanced hybrid ML architecture, and aerodynamic safety computation. Below is the implementation of the particular module.

In order to facilitate real-time prediction a powerful Repetitional state Transfer Application Programming interface (REST API) was built using Fast API and decoupled plugin architecture. Automated ML-OPS pipeline running on APSchedular ingests live METAR telemetry every 1 hour. Main cloud synchronization and real-time data broadcasting are facilitated by Superbase. That one central cloud Database where live inference results are served for the frontend dashboard. Moreover a dynamic retrain script updates model weights on a monthly basis with new ingested seasonal data through hot-reloading, thereby ensuring zero server downtime.

In order to enable the ML algorithms to find non-linear relationships in the atmosphere, the raw METAR data went through extensive feature engineering. The wind direction (0° - 360°) and time of day were converted into sine and cosine vector components by applying cyclic encoding to maintain the cycle consistency. The dew point depression was calculated by subtracting the dew point from dry bulb temperature as an indicator of air density. Furthermore, 3 hours lag features for the wind speed and pressure (QNH) along with 6 hours rolling averages were created to remove micro-anomalies and add temporal information for the models.

The Accuracy of the forecasts is maximized by utilizing an ensemble model comprising of two state-of-the-art ML frameworks. The first architecture used is Extreme Gradient Boosting (XGBoost) regressor, which provides highly accurate 1-hour nowcast by applying the gradient boosting algorithm to the lagged features of tabular data. Simultaneously, Temporal Fusion Transformer (TFT), developed with PyTorch Forecasting, generates 3-hour forecasts by finding long-term sequence dependencies in meteorological observations. The average of both predictions results in successful MAE of < 1.00 knots.

Calculation of maximum allowable crosswinds(CW) during the final approach segment [14] is necessary for aviation safety will consider that(V) is the wind speed, measured in knots, and that (q) is the angle between the true runway (R).

$$\theta = | D - R |$$

The Headwind (HW) and Crosswind (CW) are formulated as [14]:

$$HW = V \cos(\theta)$$

$$CW = V \sin(\theta)$$

For surface wind predictions by the International Civil Aviation Organization (ICAO), there should be no more than a 2.5 to 5 knots error margin [15]. Keeping the MAE value below the 2.5 knots limit, makes the system commercially viable for air traffic safety verification purposes.

Based on the experimental results, the developed hybrid model performants much better than individual models in forecasting hyperlocal aviation winds. The baseline XGBoost model produced a highly impressive MAE of 1.69 knots, but the use of deep learning through the temporal fusion transformer (TFT) lowered the MAE to about 1.00 Knot.

Finally, the combination of XGBoost and TFT into a hybrid ensemble model helped decrease the error rate below 1.00 knots. For aviation weather prediction and ATC operations, the goal criterion for high accuracy and operational safety is a mean absolute error below 2.0 knots. This proves the success of using sequential boosting algorithms together with transformer architecture to account for the short-term instability and temporal dependencies of monsoon wind in Sri Lanka.

In order to achieve robustness and prevent overfitting, hyperparameter tuning was conducted on all models. Hyperparameter tuning was done using GridsearchCV and three folds CV for XGBoost models; it was done based on negative mean absolute error. It covered essential hyperparameters such as n_estimators (300,500,1000), max_depth (5,7,9), learning rate (0.01, 0.05), and tree regularization hyperparameters subsample and colsample_bytree.

In relation to TFT, the model tuning was done by setting the learning rate at 0.03, hidden size at 32, and focus vertex at 2, with a dropout rate of 0.15 for the TFT, a lookback window of 12 hours (max encoder length = 48 intervals) was selected to allow the capturing of enough history before making predictions. In addition, the training was done using pyTorch Lightning's EarlyStopping callback, where the patience was set at three epochs for monitoring the validation loss, along sequential clipping. (0.1).

Apart from its accuracy, this study provides an MLOps framework for deploying continuously operational ML models that are scalable and suitable for deployment in production. One key challenge in weather prediction modeling is concept drift, particularly during the changes in the monsoon seasons. This problem was addressed through an automated retraining pipeline through a background job scheduler system.

The backend utilizes a decoupled modular plugin architecture based on FastAPI, which enables different weather parameters, like wind, temperature, visibility, to be processed using different API routes for high availability and fault tolerance, as one meteorological parameter forecast will not affect others in case of any failures of the particular forecast model and prevent the whole server from shutting down. The combination of a smart METAR data acquisition engine (with a 48-hour backup security net) and an interactive ATC UI calculating current attack and crosswind threat levels completes the development of the solution connecting ML theory and aviation safety in practice.

F. Cloud and Visibility Prediction

The next segment will discuss the design and structure of the Cloud and Visibility Prediction Model that that is one of the major predictive modules of the hyper-local aviation weather forecasting and verification system for VCBI. As the name implies, the major goal of this model is to make predictions of the current cloud status and visibility using the current meteorological observations retrieved from METAR observations. This type of data is highly relevant to the operation of aircraft since visibility and cloud status many influence the flight takeoff operations.

For the provision of real-time meteorological data for the prediction process, METAR data for VCBI are continually obtained from the Aviation Weather Center data source. The raw data provided by METAR are aviation-coded data and have to be preprocessed to extract the necessary for prediction meteorological variables. They include month, hour, wind direction, wind speed, dry temperature, dew point temperature, RH, QNH and present weather conditions. After processing, the data are stored in the Superbase database and serve for the storage of the observed weather data and for obtaining the most recent data when a prediction is needed.

The meteorological variable available in the dataset are then transformed to features suitable for use as inputs in the ML models. As cloud formations and visibility are affected by the combined behavior of several meteorological variables, some extra interaction features are created out of the available data points. Dew point depression feature is derived from the difference between dry temperature and dew point temperature to describe the moisture state of the atmosphere. There is also a Temperature RH interaction feature that accounts for the combine effect temperature and atmospheric moisture. A wind speed RH interaction feature is similarly created to account for the effect of wind speed on moisture state. In addition to that, a QNH-wind speed interaction feature provides more information regarding the atmospheric state. RH-squared is used to help model account for the stronger effect of high humidity levels on visibility. Weather condition is encoded as numerical values in order to incorporate weather conditions in the ML prediction process.

In order to determine which of the ML models would be more suitable for the prediction task at hand, three algorithms based on three modes were tested: Decision Tree, Random Forest, and XGBoost. The decision tree model was used first because of its simplicity and effectiveness in learning decision rules from meteorological features. RF was also used to test its performance as an ensemble algorithm DTs. XGBoost model was finally used because of its ability to learn complicated and non-linear relations between meteorological variables.

Comparison of the models tested in illustrated in the table below,

Model	Accuracy	MAE
Decision Tree	76.25%	485.1699
Random Forest	76.31%	482.9094

XG Boost	78.15%	347.7295

In this case, the improvement gained with XGBoost is especially evident with respect to the MAE metric. Even though there is a very small gap between the accuracy of the three models, XGBoost shows significantly lower prediction errors. Specifically, compared to the DT, the MAE error is lowered from 485.1699 to 347.7295, which means that the error is decreased by about 28.3%. As for RT, the MAE is decreased by about 28.0%. It can be assumed that the learn more efficiently the complex non-linear dependencies between input features and outputs.

Accordingly, XGBoost was chosen as final ML algorithm for the Cloud and Visibility prediction Module. In order to make two predictions, the module has two separate models - Cloud Status Prediction Model and Visibility Prediction Model. The Cloud model is responsible for predicting the cloud status, whereas the visibility model predicts the visibility in meters. Having two separate models allows handling the predictions based on the outpu format.

For the real-time prediction, the katest VCBI observation is received from the Superbase database when a request for prediction is made. All features are preprocessed using the same procedure that was used during the model training. Then the future vector is supplied to the XGBoost models, which make predictions and send them through the backend API. Thus the same feature structure is preserved for both the training and prediction phase.

The backend prediction mechanism of the proposed module is realize by using a Flask API where it gets the needed meteorological parameters and then predicts using the trained XGBoost model. Here, the output of cloud parameters and then predicts using the trained XGBoost model. Here the output of the cloud parameter is mapped to the corresponding cloud parameter is mapped to the corresponding visivility.

Prediction process overall could be explained as follows: METAR Data → Data Parsing → Superbase Database → Features Generation → XGBoost Models → Cloud & Visibility Prediction → Flask API → Dashboard

Prediction result is displayed on the dashboard in human-readable from, which minimizes the manual work of interpreting aviation weather codes. From the experimental results, XGBoost was found to outperform other models with 78.15% accuracy and 347.7295 of MAE, beating DT and RF model.

G. Qnh, Dewpoint and Rh Prediction

This part of the system that predicts QNH, Dewpoint and RH uses a data-driven method. It relies on the Light Gradient Boosting Machine (LightGBM) and the August-Roche-Mangus formula from thermodynamics. I have been that this combination gives a balance between statical learning and physical reality.

The system keeps pilling local weather data from a central SQLite database. To make sure the time order stays correct all records are sorted by UTC timestamps. When a value is missing the system copies the known value forward so that no gaps appear.

To make QNH, Dewpoint and RH predictions more accurate many features are created from observations. The hour of the day is turned into sine and cosine value so that the daily cycle shows up in the model. For Dewpoint the month of the year is encoded in the way to capture seasonal changes. That also add the temperature spread - the different between air temperature and Dewpoint - because it tells the model how moist the air is. Lagged values and QNH momentum features capture how the atmosphere changes over time. Short-term trends are seen with 3-hour and 6-hour rolling averages.

The LightGBM model is trained separately to forecast QNH and Dewpoint three hours ahead. During training the target values are shifted three hours. The data is split chronologically: 80% for training and 20% for testing with no shuffling. This keeps observations hidden from the model. The Dewpoint model reached an R^2 of 0.66 and MAE of around 0.58 °C. The QNH model achieved an absolute error of roughly 1.8 hpa.

Unlike QNH and Dewpoint RH is not predicted by its ML model. Instead the predicted dewpoint and predicted temperature are fed into the August-Roche-Magnus equation to obtain future RH:

$$RH = 100 \times \exp\left[\left(\frac{17.67 T_d}{T_d + 243.5}\right) - \left(\frac{17.67 T}{T + 243.5}\right)\right]$$

Where the symbol (T_d) stands for the predicted Dewpoint temperature (°C) and the symbol (T) stands for the predicted dry temperature (°C). This hybrid approach keeps the predictions of QNH, Dewpoint and RH consistent.

Finally the forecasting subsystem is tied into an automated MLOps pipeline that regularly pulls meteorological observations and retrains the models. Also added a real-time verification step that compares the predictions, with ground-truth data and tracks model performance over time. This setup allows the system to keep delivering reliable three-hour-ahead forecasts of QNH, dewpoint and RH when local or monsoon weather changes.

H. Aviation Physics and Hazard Analysis

Forecast QNH and temperature are converted to pressure altitude, ISA temperature, and DA. DA is subsequently categorized as either NORMAL, ELEVATED, or CRITICAL based on the airport specific criteria set out in the system. A radiation fog forecast criterion is also employed on the basis of the VCBI experience with respect to past. The high-risk criterion is satisfied if there is a decrease in temperature of 1.0°C within three hours and if the QNH is equal to or greater than 1012.0 hPa with a variation of 0.5 hPa.

I. Forecast Verification and Evaluation

The generated forecasts are saved along with their target timestamps. As soon as the future observation is made, the validation stage finds the forecast to match with the observation based on the METAR report and determines the error in the forecast. The metrics MAE, RMSE, and R^2 are used for regression analysis. For cloud classification, accuracy metrics are reported. The verification stage is done independently for each weather variable.

J. MLOps and Continuous Traning

Model maintenance is included to minimize the impact of change in weather behavior and concept drift. Background tasks are performed to get updated METAR observations and perform database synchronization. Temperature pressure

system includes automated monthly training, on demand training using a FastAPI call, and hot reloading of updated RF and LSTM models. Wind subsystem includes periodic collection of data, database synchronization, and monthly traning. Cloud and visibility subsystem updates its XGBoost models by retraning based on collected observations,whereas the QNH and dew point subsytem includes automatic LightGBM retraining and live validation. This helps in incorporating observed monsoon behaviors into the forecasting model.

K. Sytem Intergration

FastAPI delivers backend capabilities for the retrieval of real time data, inference, managing the models, and re-training. Superbase combined with SQLAlchemy is used to store observations, predictions, and verification logs. React interfaces are used to visualize real time weather, predictions, aviation parameters, and warning. the cloud and visibility module also has a Streamlit interface. The output combines the forecasts with aviation performance and hazards.

IV. RESULT AND DISCUSSION

The experiment were performed with the models developed for VCBI to evaluate the effectiveness of the proposed local aviation weather forecasting and validation system. The assessment of the system included the main meteorology parameters needed such as temperature, pressure, wind, clouds, visibility, and dew point. Since each of the meteorological parameters has a different time behavior and non-linearity, a variety of ML and DL structures were evaluated.

However, it was found in the prediction test of temperature and pressure that different leaning approaches can increase the accuracy of short-term prediction when all of them are combined. While the isolated RF model was able to obtain an R^2 score of about 68% and LSTM model about 25%, the hybrid RF-LSTM approach yielded an R^2 score of about 84% with an MAE score of about 0.86 according to the described evaluation scenario. This indicates that such a combination of the two learning styles may generate a more accurate model of the short-term atmospheric state. These "calculated" values of temperature and QNH are then used in the aviation physics section to calculate the PA, ISA temperature, and DA.

During the wind forecasting experiments it was once again found that the choice of model should be based on the forecasting horizon. For the 1 hour nowcasting, the MAE obtained by XGBoost was found to be 1.69 knots. The implementation of TFT and the hybrid ensemble method between TFT and XGBoost led to an MAE of less than 1.00 knot. These results show that the attention based method for temporal modeling can be helpful to improve the rapid nonlinear learning capability of the XGBoost model on longer forecasting time horizon. Also, the output of the wind is converted to the headwind and crosswind component values measured along the runway.

Cloud cover, and visibility prediction tasks were both undertaken using Decision Tree, RF and XGBoost algorithms. Cloud prediction accuracy of Decision Tree was 76.25%, RF showed better results, namely,76.31%, the best results were produced by XGBoost , 78.15% from prediction and. it can be seen from the results that the use of the sequential gradient boosting helped to learn nonlinear dependencies better. At the same time, the results show that the prediction of cloud cover and visibility is a very challenging task because of their rapid changes and convective rainfall due to the monsoon.

If LightGBM is used to forecast QNH and dewpoint, an R^2 of ~0.66 and a MAE of ~0.58°C is acheived for the dew point forecast. In the meanwhile, the MAE for the prediction of QNH was about 1.8 hPa. The dew point and predicted temperature were used in the August-Roche-Magnus formula to calculate RH. It eliminates the need for another RH forecasting model without losing interdependency of the temperature, dew point, and RH values.

Overall, the performance of the forecast system is related to the parameters forecasted and the time period of the forecasts. A hybrid architecture of RF and LSTM model was obtained the best forecast quality for the temperature and QNH parameters, and for the wind forecast the best was obtained with a hybrid architecture of XGBoost and TFT model, while for the cloud and visibility parameters, the best was obtained with XGBoost model and for the dew point forecast, the LightGBM model was applied.

The outputs of the forecasts were also combined with the verification and the aviation decision-support layer. Errors in the prediction can be compared at the airport and expressed in terms of the difference between the predicted and observed values in a METAR. the system also converts meteorological forecasts into actionable data, like DA, runway headwind and crosswind, temperature, pressure, visibility and

thermodynamic hazard warnings

The other advantage of the training and validation process is the opportunity to use the data collected in the METAR stations in future upgrades of the models. This is particularly so in forecasting for the monsoon region because the behavior of the atmosphere varies with time. It is important to mention that the results that have been obtained are based on the datasets used in the evaluation process. More tests to be conducted for other seasons of the monsoon and with other sets of data for airports.

V. SYSTEM IMPLEMENTATION

The system was built as a platform to forecast and verify aviation weather at VCBI. The system pulls METAR data runs ML predictions manages a database computes aviation parameters verifies forecasts and offers an interactive dashboard.

Live METAR observations are fetched from the Aviation Weather Center API. Turned into clear weather variables. The system saves these processed observations in a Superbase database through Superbase so it can keep both the data and past prediction records.

The AI layer has four forecasting parts. The temperature and QNH part uses a RF-LSTM model. The wind part mixes XGBoost and TFT models. Cloud cover and visibility are predicted with XGBoost. QNH and dewpoint are predicted with LightGBM. The system loads these trained models in the backend to produce short-term forecasts from the observations.

the backend runs on FastAPI and supplies prediction, data retrieval, verification and model management services. Background tasks keep collecting METAR data and retraining models. The system also calculates aviation values like DA, headwind, crosswind, and thermodynamic hazard indicators, from the predicted weather.

The frontend dashboard shows users live observations, forecast numbers, aviation values, verification results, and warning signs. The main dashboard is built with React.js while a Streamlit interface displays cloud and visibility data. Thus,

the system brings all the AI models together into one airport-level aviation weather decision support system.

VI. Conclusion

In this research, a novel Hyperlocal aviation weather forecasting and verification system was proposed to be used at VCBI. Such a system will provide an opportunity to create accurate and timely airport-level weather information considering the atmosphere conditions in the area of interest. For main components including temperature and atmospheric pressure, wind, cloud and visibility, and humidity-related parameters were implemented in this research. In addition, different ML algorithms were used depending on the characteristics of the considered weather parameter that allowed capturing linear and non-linear relationships in the atmosphere. For the creation of the created forecasts are checked against future observations and the relevant TAF reports to validate the reliability of the predictions.

Another crucial aspect of this system is the possibility of integrating weather forecasting with aviation-oriented interpretation. It means that the raw meteorological information is converted to the useful one for aviation purposes, for example, density altitude calculation, runway-oriented wind components, visibility analysis, etc.

Another advantage of this system is its MLOps architecture that provides a possibility of automated data collection and management, models' retraining, validation, and deployment. Using the interaction of backend services, databases, machine learning models, and dashboards, the proposed system implements a complete workflow from the process of obtaining the data up to verification of the created forecast.

The proposed system allows combining AI, aviation meteorology, real-time data processing, physical aviation calculations, forecast verification, and MLOps to improve Hyper-local weather forecasting at the airport. The system developed in this research will provide an opportunity to have better weather awareness and operational decision-making at VCBI. Which is not possible using conventional regional weather forecasting techniques.

For further work, the scope of historical and real-time datasets can be expanded, the possibility of rapidly changing weather events can be improved, extended validation can be conducted considering different monsoons and inter-monsoon periods. In addition, the use of other meteorological and remote sensing data can be implemented. Another possible direction is the application of the proposed framework to another in Sri Lanka.

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VIII. Reference

[1] H. R. K. Nagahamulla, U. R. Ratnayake and A. Ratnaweera, "Monsoon rainfall forecasting in Sri Lanka using artificial neural networks," in *6th International Conference on Industrial and Information Systems (ICIIS)*, Kandy, Sri Lanka, 2011.

[2] K. Hennayake, R. Dinalankara and D. Y. Mudunkotuwa, "Machine Learning Based Weather Prediction Model for Short Term Weather Prediction in Sri Lanka," in *2021 10th International Conference on Information and Automation for Sustainability (ICIAfS)*, Negombo, Sri Lanka, 2021.

[3] U. D. Maduranga and M. Ediridinghe, "Climatology of Lightning Activities over Bandaranaike International Airport in Sri Lanka," *International Journal of Sustainable Development and Planning*, vol. 16, no. 6, p. 1027–1038, 31 October 2021.

[4] I. Gultepe, R. Sharman, P. D. Williams, B. Zhou, G. Ellrod, P. Minnis, S. Trier, S. Griffin, S. Yum, B. Gharabaghi, W. Feltz, M. Temimi, Z. Pu, L. Storer, P. Kneringer, M. Weston, H.-y. Chuang, L. Thobois, A. Dimri, S. Dietz, G. B. França, M. Almeida and F. Albquerque Neto, "A Review of High Impact Weather for Aviation Meteorology," *Pure and Applied Geophysics*, vol. 176, p. 1869–1921, 6 May 2019.

[5] K. Oo and K. Oo, "Analysis of the Most Common Aviation Weather Hazard and Its Key Mechanisms over the Yangon Flight Information Region," *Advances in Meteorology*, vol. 2022, pp. 1-15, 2022.

[6] D. Sládek, "AI framework for automated terminal aerodrome forecasting," *Aviation*, vol. 29, no. 4, pp. 210-221, 26 November 2025.

[7] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-k. Wong and W.-c. Woo, "Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting," in *Advances in Neural Information Processing Systems 28 (NIPS 2015)*, Montreal, Canada, 2015.

[8] S. K. Sharma, D. K. Pandiya, V. R. Joshi, P. Bansal and D. K. Sharma, "Artificial Intelligence for Real-Time Weather Forecasting Using Neural Networks," in *Intelligent Systems Using Semiconductors for Robotics and IoT*, CRC Press,

2025, p. 347–353.

[9] P. D. Dueben and P. Bauer, "Challenges and design choices for global weather and climate models based on machine learning," *Geoscientific Model Development*, vol. 11, no. 10, pp. 10.5194/gmd-11-3999-2018, 2018.

[10] S. Rasp and S. Lerch, "Neural Networks for Postprocessing Ensemble Weather Forecasts," *Monthly Weather Review*, vol. 146, no. 11, p. 3885–3900, 2018.

[11] P. Hu, J. Zhang and N. Li, "Research on Flight Delay Prediction Based on Random Forest," in *2021 IEEE 3rd International Conference on Civil Aviation Safety and Information Technology (ICCASIT)*, Changsha, China, 2021.

[12] P. Boonyuen, F. Wu, P. Phunthirawuth** and Y. Zhao, "Impact of GPS RO and Radiance Data Assimilation on Numerical Weather Prediction," in *2016 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Beijing, China, 2016.

[13] S. Asifa, B. M. Samarsitha, R. Neha, R. N. Jalagam and G. Saranya, "Lightning Risk Prediction for Aviation Safety," in *2025 1st International Conference on Radio Frequency Communication and Networks (RFCoN)*, 2025.

[14] Federal Aviation Administration, "Pilot’s Handbook of Aeronautical Knowledge," U.S. Department of Transportation, Federal Aviation Administration, 2023.

[15] International Civil Aviation Organization, "Meteorological Service for International Air Navigation," International Civil Aviation Organization (ICAO), 2018.